

module-09-subtraction-nikhilam - Darshana (60-min workshop)

IKS Curriculum - senior band

Module 9 — Subtraction by Nikhilam (Darśana)

Sanskrit: *Nikhilam Navataścaramam Daśataḥ* · **Band:** Senior (9-12) · **Duration:** 60 min ·
Path: Darśana (Workshop)

“All from 9, last from 10” — a complement method for subtraction.

Darśana note. This workshop condenses Module 9’s full 10-day arc into one hour. For the deeper version — daily lesson plans, quizzes, assessments, slide decks — switch to [Adhyayana](#) (the Full Course path).

Opening hook · 5 min

Project: $10000 - 1547$. Long subtraction takes ~30 seconds. “*Nikhilam takes ~5 seconds. Today: the method + its bridge to TWO’S COMPLEMENT in modern computing.*”

Core idea · 15 min

Nikhilam complement method (Tirthaji, 1965, Sūtra 2).

For $B - N$ (where B is 10^n): - Identity: $B - N = (B - 1) - N + 1$ - Procedure: “all from 9” gives $(B - 1) - N$ digit-wise (no borrows because every digit of $(B - 1)$ is 9) - “+1” gives the last-digit-from-10 correction

Modern parallel: two’s complement in binary computing.

To compute $A - B$ in binary, computers don’t have a subtraction circuit. Instead: 1. Invert (one’s complement) every bit of B . 2. Add 1 (two’s complement). 3. Add A .

Same identity, in base 2. **Nikhilam = two’s complement, in base 10.** The principle is the same.

This is a genuine insight, not a forced parallel.

Demonstration · 10 min

Binary bridge. On the board, compute $9 - 5$ in binary using two’s complement. Then compute $10 - 7$ in decimal using Nikhilam. Highlight: same identity, two bases.

Source moment · 5 min

Quoted reference. Tirthaji, *Vedic Mathematics* (1965), Sūtra 2:

Nikhilam Navataścaramam Daśataḥ

“All-from-nine and last-from-ten.”

The sūtra name is Sanskrit. The technique it labels is the decimal complement, which is well-attested across pre-modern arithmetic traditions (Chinese, Arabic, European). Tirthaji’s contribution: organising under Sanskrit sūtra-names.

Hands-on activity · 15 min

Computer-subtraction explainer poster. Each student designs a poster showing: (a) Nikhilam decimal subtraction with one example, (b) two’s complement binary subtraction with the same number in binary. Display side-by-side.

Reflection + take-home · 10 min

“Argue: which deserves more curriculum time — the Nikhilam method or its connection to two’s complement computer arithmetic? Defend your answer.”

Go deeper into Adhyayana

The 10 days you’re skipping if you only do this workshop:

- **Day 01** — The sūtra — all from 9, last from 10
- **Day 02** — Complement to 1000
- **Day 03** — Subtract from a power of 10
- **Day 04** — General subtraction
- **Day 05** — The algebra — why it works
- **Day 06** — Bar model proof
- **Day 07** — Two’s complement in binary
- **Day 08** — Speed practice
- **Day 09** — Mixed practice
- **Day 10** — Final assessment + computer-subtraction poster

→ [Open the Adhyayana track for this module](#) to access all 10 days, the 4-audience PDFs, the slide deck, the quizzes, and the project rubric.